

Lukas Frotton

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SUMMARY

- Computer Science undergraduate with a strong foundation in algorithms, data structures, and mathematical reasoning. Experienced in backend systems, databases, and machine learning, with growing interest in research-oriented software development.

EDUCATION

Rutgers University

New Brunswick, NJ

B.S. in Computer Science; Minor in Mathematics

Expected May 2026

- **Relevant Coursework:** Linear Algebra, Discrete Structures II, Systems Programming, Differential Equations, Graph Theory, Computational Robotics, Design and Analysis of Algorithms, Intro to Deep Learning, Theory of Complex Valued Functions, Elementary Partial Differential Equations

Montclair State University

Montclair, NJ

Completed Coursework toward B.S. (Transferred)

Sep 2022 – Dec 2023

- **Relevant Coursework:** Computer Systems, Discrete Mathematics

EXPERIENCE

Research Assistant

Sep 2025 – Present (8–20 hrs/week)

Dr. Yu Lab, Rutgers University

- Worked on the MAP research team developing an application intended to support pain monitoring.
- Built a hipaa compliant accessible user authentication system.
- Contributed to software development tasks in collaboration with faculty and research team members.

Data Reporting Analyst

Aug 2024 – Present (3–10 hrs/week)

Reworld / Robert Half

- Analyzed and validated operational data using **NetSuite** and automated Salesforce reporting workflows to improve accuracy and efficiency.

STEM Teacher / Mentor

Mar 2023 – Aug 2023

Pathways for Exceptional Children

- Led STEM workshops introducing **Scratch** and **Python**; trained and coordinated student mentors.

PROJECTS

Complex-Valued Neural Networks (CVNN) Research

Ongoing

Independent Research

- Conducting research on complex-valued neural network architectures, focusing on training behavior and representation.
- Implementing experimental models and training pipelines in **Python** using numerical and deep learning libraries.

ProfessorRank

Web Application Project

- Built and deployed a scalable web platform using **Next.js** and **Prisma ORM**, supporting over 100 concurrent users.

MapRootDB

Systems Programming Project

- Developing an in-memory database in **Rust** using a hybrid graph-tree structure with emphasis on efficient data access.

The Diabeater

Machine Learning Project

- Implemented a diabetes risk prediction application using **TensorFlow** and **PyTorch**.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, Rust, SQL

Frameworks: React, Next.js, TensorFlow, PyTorch, FastAPI, Flask

Databases: PostgreSQL, MongoDB, SQLite

Tools: Git, Linux, Docker, AWS, Google Cloud